Lab04

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Laboratório 4

Q.1 Carregue os pacotes readxl e tidyverse.

```
# Carregando os pacotes
library(readxl)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.3     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

Q.2 Crie uma variável chamada fname que contenha o caminho para o arquivo WDIEXCEL.xlsx, disponível no seu diretório de dados. Confirme que o R identifica a existência do arquivo utilizando o comando file.exists

```
# Variável com o caminho absoluto
fname <- "WDIEXCEL.xlsx"

# Verificando se o arquivo existe
file.exists("WDIEXCEL.xlsx")
```

```
[1] TRUE
```

```r
file.exists(fname)
```

```
[1] TRUE
```

Q.3 Leia a planilha "Country" e armazene no objeto countries.

```r
excel_sheets("WDIEXCEL.xlsx")
```

```
[1] "Data"           "Country"        "Series"         "country-series"
[5] "series-time"    "footnote"
```

```r
countries <- read_excel(fname, sheet = "Country")
countries
```

```
# A tibble: 264 x 31
   `Country Code` `Short Name`            `Table Name` `Long Name` `2-alpha code`
   <chr>          <chr>                   <chr>        <chr>       <chr>
 1 ABW            Aruba                   Aruba        Aruba       AW
 2 AFE            Africa Eastern and So~ Africa East~ Africa Eas~ ZH
 3 AFG            Afghanistan             Afghanistan  Islamic St~ AF
 4 AFW            Africa Western and Ce~ Africa West~ Africa Wes~ ZI
 5 AGO            Angola                  Angola       People's R~ AO
 6 ALB            Albania                 Albania      Republic o~ AL
 7 AND            Andorra                 Andorra      Principali~ AD
 8 ARB            Arab World              Arab World   Arab World  1A
 9 ARE            United Arab Emirates    United Arab~ United Ara~ AE
10 ARG            Argentina               Argentina    Argentine ~ AR
# i 254 more rows
# i 26 more variables: `Currency Unit` <chr>, `Special Notes` <chr>,
#   Region <chr>, `Income Group` <chr>, `WB-2 code` <chr>,
#   `National accounts base year` <chr>,
#   `National accounts reference year` <dbl>, `SNA price valuation` <chr>,
#   `Lending category` <chr>, `Other groups` <chr>,
#   `System of National Accounts` <chr>, ...
```

Q.4 Mantenha apenas os registros de 5 países (Brasil e outros 4 de sua escolha).

```
countries <-  countries |>
  filter(`Short Name` %in% c("Brazil", "China", "Russian Federation", "United States", "India

countries
```

```
# A tibble: 5 x 31
  `Country Code` `Short Name`       `Table Name`      `Long Name` `2-alpha code`
  <chr>          <chr>              <chr>             <chr>       <chr>
1 BRA            Brazil             Brazil            Federative~ BR
2 CHN            China              China             People's R~ CN
3 IND            India              India             Republic o~ IN
4 RUS            Russian Federation Russian Federati~ Russian Fe~ RU
5 USA            United States      United States     United Sta~ US
# i 26 more variables: `Currency Unit` <chr>, `Special Notes` <chr>,
#   Region <chr>, `Income Group` <chr>, `WB-2 code` <chr>,
#   `National accounts base year` <chr>,
#   `National accounts reference year` <dbl>, `SNA price valuation` <chr>,
#   `Lending category` <chr>, `Other groups` <chr>,
#   `System of National Accounts` <chr>, `Alternative conversion factor` <lgl>,
#   `PPP survey year` <lgl>, `Balance of Payments Manual in use` <chr>, ...
```

Q.5 Leia a planilha "Data" e armazene no objeto WDI.

```
WDI <- read_excel(fname, sheet = "Data")
WDI
```

```
# A tibble: 396,970 x 70
   `Country Name` `Country Code` `Indicator Name` `Indicator Code` `1960` `1961`
   <chr>          <chr>          <chr>            <chr>             <dbl>  <dbl>
 1 Africa Easter~ AFE            Access to clean~ EG.CFT.ACCS.ZS       NA     NA
 2 Africa Easter~ AFE            Access to clean~ EG.CFT.ACCS.RU.~     NA     NA
 3 Africa Easter~ AFE            Access to clean~ EG.CFT.ACCS.UR.~     NA     NA
 4 Africa Easter~ AFE            Access to elect~ EG.ELC.ACCS.ZS       NA     NA
 5 Africa Easter~ AFE            Access to elect~ EG.ELC.ACCS.RU.~     NA     NA
 6 Africa Easter~ AFE            Access to elect~ EG.ELC.ACCS.UR.~     NA     NA
 7 Africa Easter~ AFE            Account ownersh~ FX.OWN.TOTL.ZS       NA     NA
 8 Africa Easter~ AFE            Account ownersh~ FX.OWN.TOTL.FE.~     NA     NA
```

```
 9 Africa Easter~ AFE            Account ownersh~ FX.OWN.TOTL.MA.~     NA     NA
10 Africa Easter~ AFE            Account ownersh~ FX.OWN.TOTL.OL.~     NA     NA
# i 396,960 more rows
# i 64 more variables: `1962` <dbl>, `1963` <dbl>, `1964` <dbl>, `1965` <dbl>,
#   `1966` <dbl>, `1967` <dbl>, `1968` <dbl>, `1969` <dbl>, `1970` <dbl>,
#   `1971` <dbl>, `1972` <dbl>, `1973` <dbl>, `1974` <dbl>, `1975` <dbl>,
#   `1976` <dbl>, `1977` <dbl>, `1978` <dbl>, `1979` <dbl>, `1980` <dbl>,
#   `1981` <dbl>, `1982` <dbl>, `1983` <dbl>, `1984` <dbl>, `1985` <dbl>,
#   `1986` <dbl>, `1987` <dbl>, `1988` <dbl>, `1989` <dbl>, `1990` <dbl>, ...
```

Q.6 Selecione apenas as colunas:

- Country Code

- Indicator Code

- E todos os anos de 1960-2018

```r
WDI <- WDI[, c(2, 4:63)]
WDI
```

```
# A tibble: 396,970 x 61
   `Country Code` `Indicator Code`  `1960` `1961` `1962` `1963` `1964` `1965`
   <chr>          <chr>              <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>
 1 AFE            EG.CFT.ACCS.ZS        NA     NA     NA     NA     NA     NA
 2 AFE            EG.CFT.ACCS.RU.ZS     NA     NA     NA     NA     NA     NA
 3 AFE            EG.CFT.ACCS.UR.ZS     NA     NA     NA     NA     NA     NA
 4 AFE            EG.ELC.ACCS.ZS        NA     NA     NA     NA     NA     NA
 5 AFE            EG.ELC.ACCS.RU.ZS     NA     NA     NA     NA     NA     NA
 6 AFE            EG.ELC.ACCS.UR.ZS     NA     NA     NA     NA     NA     NA
 7 AFE            FX.OWN.TOTL.ZS        NA     NA     NA     NA     NA     NA
 8 AFE            FX.OWN.TOTL.FE.ZS     NA     NA     NA     NA     NA     NA
 9 AFE            FX.OWN.TOTL.MA.ZS     NA     NA     NA     NA     NA     NA
10 AFE            FX.OWN.TOTL.OL.ZS     NA     NA     NA     NA     NA     NA
# i 396,960 more rows
# i 53 more variables: `1966` <dbl>, `1967` <dbl>, `1968` <dbl>, `1969` <dbl>,
#   `1970` <dbl>, `1971` <dbl>, `1972` <dbl>, `1973` <dbl>, `1974` <dbl>,
#   `1975` <dbl>, `1976` <dbl>, `1977` <dbl>, `1978` <dbl>, `1979` <dbl>,
#   `1980` <dbl>, `1981` <dbl>, `1982` <dbl>, `1983` <dbl>, `1984` <dbl>,
#   `1985` <dbl>, `1986` <dbl>, `1987` <dbl>, `1988` <dbl>, `1989` <dbl>,
#   `1990` <dbl>, `1991` <dbl>, `1992` <dbl>, `1993` <dbl>, `1994` <dbl>, ...
```

Q.7 Confirme se os dados wm WDI estão em formato tidy. Se não estiverem, transforme-os para tal formato. Faça de forma que exista uma coluna referente ao ano chamada YEAR.

```
# Temos que os dados não estão em formato tidy, pois a variável anos está dividida em várias

# Colocando WDI em formato tidy
WDI <- WDI |>
  pivot_longer(
    cols = 3:61,
    names_to = "YEAR",
    values_to = "DATA"
  ) |>
  mutate(YEAR = as.integer(YEAR))
WDI
```

```
# A tibble: 23,421,230 x 4
   `Country Code` `Indicator Code`  YEAR  DATA
   <chr>          <chr>            <int> <dbl>
 1 AFE            EG.CFT.ACCS.ZS    1960    NA
 2 AFE            EG.CFT.ACCS.ZS    1961    NA
 3 AFE            EG.CFT.ACCS.ZS    1962    NA
 4 AFE            EG.CFT.ACCS.ZS    1963    NA
 5 AFE            EG.CFT.ACCS.ZS    1964    NA
 6 AFE            EG.CFT.ACCS.ZS    1965    NA
 7 AFE            EG.CFT.ACCS.ZS    1966    NA
 8 AFE            EG.CFT.ACCS.ZS    1967    NA
 9 AFE            EG.CFT.ACCS.ZS    1968    NA
10 AFE            EG.CFT.ACCS.ZS    1969    NA
# i 23,421,220 more rows
```

Q.8 Confirme se a coluna referente ao ano está em formato inteiro.

```
# Temos que a coluna YEAR é composta de inteiros
glimpse(WDI)
```

```
Rows: 23,421,230
Columns: 4
$ `Country Code`   <chr> "AFE", "AFE", "AFE", "AFE", "AFE", "AFE", "AFE", "AFE~
```

```
$ `Indicator Code` <chr> "EG.CFT.ACCS.ZS", "EG.CFT.ACCS.ZS", "EG.CFT.ACCS.ZS",~
$ YEAR             <int> 1960, 1961, 1962, 1963, 1964, 1965, 1966, 1967, 1968,~
$ DATA             <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
```

```
is.integer(WDI$YEAR)
```

```
[1] TRUE
```

Q.9 Para o objeto WDI, mantenha apenas os registros dos países listados em countries. Selecione as colunas:

- Country Code

- Short Name

- YEAR

- SP.DYN.CBRT.IN

```
WDI <- WDI |>
  filter(`Country Code` %in% countries$`Country Code`) |>
  filter(`Indicator Code` == "SP.DYN.CBRT.IN") |>
  left_join(
    countries |> select(`Country Code`, `Short Name`),
    by = "Country Code"
  ) |>
  select(`Country Code`, `Short Name`, YEAR, DATA)
WDI
```

```
# A tibble: 295 x 4
   `Country Code` `Short Name`  YEAR  DATA
   <chr>          <chr>        <int> <dbl>
 1 BRA            Brazil        1960  43.8
 2 BRA            Brazil        1961  43.3
 3 BRA            Brazil        1962  42.7
 4 BRA            Brazil        1963  42.0
 5 BRA            Brazil        1964  41.0
 6 BRA            Brazil        1965  39.8
 7 BRA            Brazil        1966  38.6
```

```
 8 BRA            Brazil        1967  37.4
 9 BRA            Brazil        1968  36.3
10 BRA            Brazil        1969  35.7
# i 285 more rows
```

Q.10 Ordene o objeto de forma que, para cada país, as informações estejam ordenadas por ano.

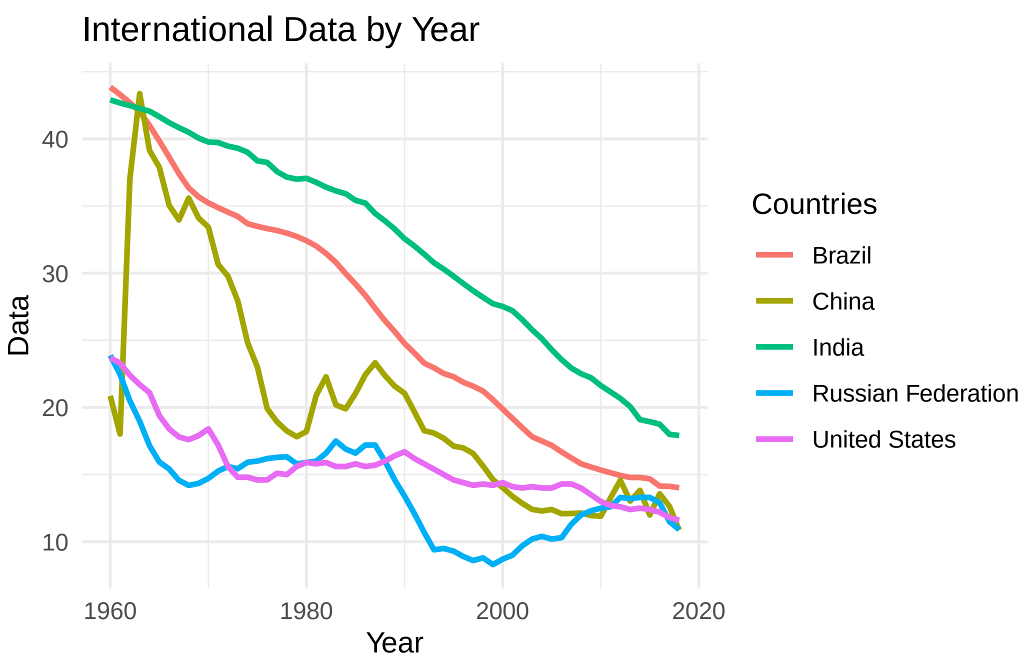
```
WDI <- WDI |>
  arrange(`Country Code`, `YEAR`)
WDI
```

```
# A tibble: 295 x 4
   `Country Code` `Short Name`  YEAR  DATA
   <chr>          <chr>        <int> <dbl>
 1 BRA            Brazil        1960  43.8
 2 BRA            Brazil        1961  43.3
 3 BRA            Brazil        1962  42.7
 4 BRA            Brazil        1963  42.0
 5 BRA            Brazil        1964  41.0
 6 BRA            Brazil        1965  39.8
 7 BRA            Brazil        1966  38.6
 8 BRA            Brazil        1967  37.4
 9 BRA            Brazil        1968  36.3
10 BRA            Brazil        1969  35.7
# i 285 more rows
```

Q.11 Construa um gráfico de linhas, usando o pacote ggplot2, que tenha no eixo X a variável YEAR, no eixo Y a variável SP.DYN.CBRT.IN e que cada linha corresponda a um país diferente.

```
ggplot(WDI, aes(x = YEAR, y = DATA, group = `Short Name`, color = `Short Name`)) +
  geom_line(linewidth = 1) +
  labs(
    title = "International Data by Year",
    x = "Year",
    y = "Data",
    color = "Countries"
```

```r
  ) +
  theme_minimal()
```



Tempo

```r
Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()
```

```
[1] "2026-09-09 17:31:46 -03"
```